

Neuroscientific Frontiers — Module 03: Perception

— Study Questions

1. What are the three main oscillatory frequency bands in predictive coding? What signal does each carry and in which cortical layers is each generated?
2. Why is gamma associated with feedforward (prediction error) signals? What anatomical evidence supports this mapping?
3. Why is beta associated with feedback (prediction) signals? What happens to beta power when a prediction is confirmed vs. violated?
4. How does alpha implement precision gating (attention)? Why does alpha power *increase* over task-irrelevant cortex rather than decrease?
5. Explain the mismatch negativity (MMN) in oscillatory terms. What frequency band increases during the deviant response? How does this connect to the 101 treatment of MMN?
6. How does ketamine (NMDA blockade) provide pharmacological evidence for the feedforward/feedback distinction? What specific oscillatory changes does it produce?
7. What do binocular rivalry studies reveal about the oscillatory dynamics of perception? How do competing percepts relate to competing generative models?
8. Describe the laminar MEG (Michalareas et al., 2016) findings. How do they provide direct evidence for the oscillatory predictive coding framework?
9. How does spatial attention modulate prediction error precision? What oscillatory changes occur over attended vs. unattended cortical regions?
10. Explain the attentional blink as temporary precision depletion. How does this differ from the traditional "bottleneck" explanation?
11. What is phase-amplitude coupling (PAC)? How does it implement hierarchical control of prediction errors by higher-level predictions?
12. How does surround suppression in V1 demonstrate prediction error computation? Why are expected stimuli within the receptive field context suppressed?
13. What is stimulus-specific adaptation (SSA) in auditory cortex? How does it relate to the mismatch negativity and predictive coding?
14. Explain sensory attenuation of self-generated touch in oscillatory terms. What predictions are generated and what errors are suppressed?

15. How do temporal attention and delta-theta oscillations coordinate precision allocation to specific moments in time?
16. What is the relationship between feature attention and the precision weighting of feature-specific prediction error neurons?
17. Connect the oscillatory framework to anesthesia: how do changes in oscillatory dynamics during anesthetic-induced unconsciousness relate to precision collapse?
18. How do cross-modal prediction errors (e.g., audiovisual mismatch) differ from within-modal errors? What oscillatory signatures distinguish them?
19. How does the oscillatory framework explain why chronic pain persists even without ongoing tissue damage? Is chronic pain a precision disorder?
20. Critically evaluate: Does the gamma-for-errors, beta-for-predictions hypothesis hold universally, or are there important exceptions? What would falsify this framework?